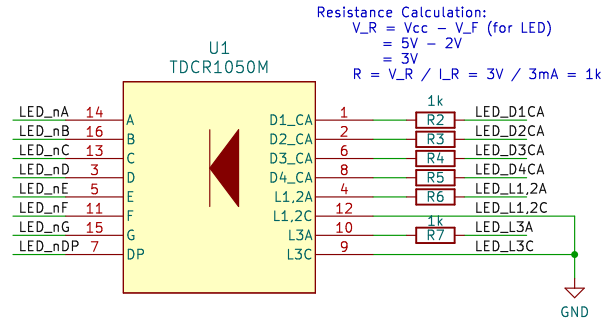


HKN DIY Watch Project

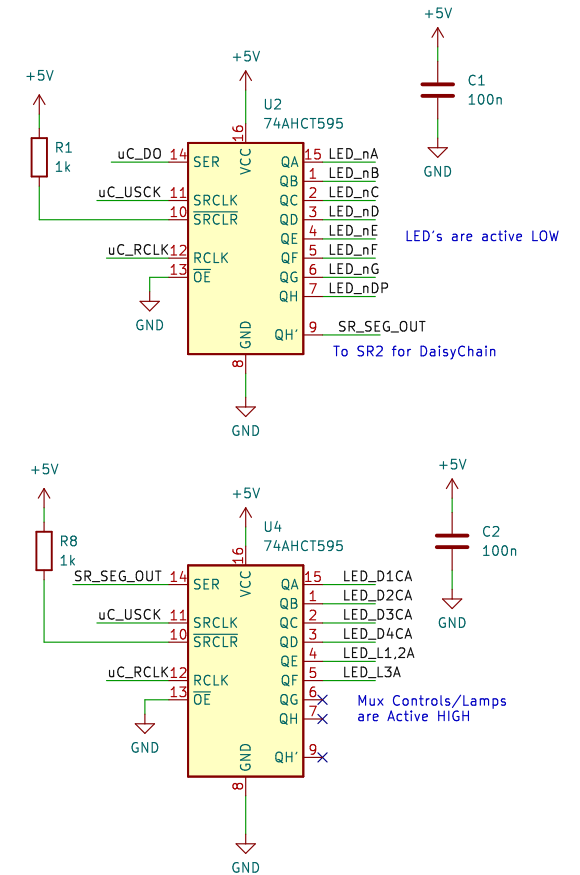
By: John Mentgen

A through-hole DIY watch kit that allows you to make your own wristwatch!

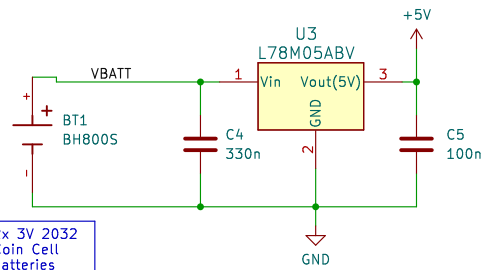
4 Digit 7 Segment Display



Shift Registers For LED Multiplexing



Battery & Voltage Regulator



Capacitor values as supplied in datasheet

2x 3V 2032
Coin Cell
batteries

Ok, so after some rough calculations, I have found the following current draw:

Total Current Draw = $I_{UC} + I_{LEDs} + I_{LDO} + I_{SRs}$

$I_{UC} = 8mA$ when active (from datasheet)

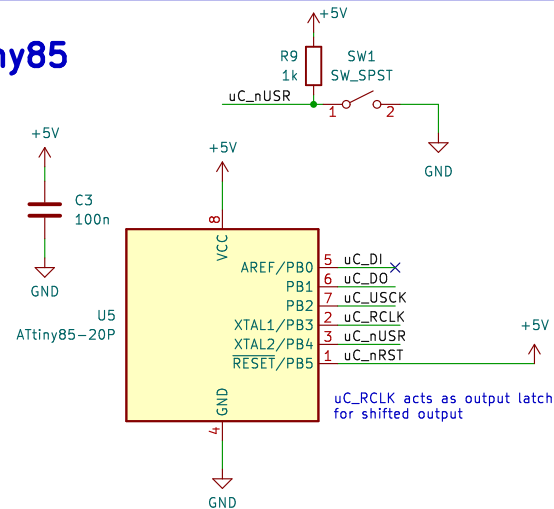
$I_{LEDs} = [3mA \text{ per LED}] * [8 \text{ LEDs/Seg}] * [1 \text{ seg}] + [9mA \text{ for 3 Lamps}] = 33 \text{ mA}$

$I_{LDO} = 6mA$ (typical from datasheet)

$I_{SRs} = 0mA$ (negligible)

This adds to 47mA. For 2 2032 Coin Cells, they have 470mAh. Under continuous use, this watch can stay alive for 10 hours. Now, how continuous that is will depend on user use, but we'll see how that fairs once its built.

ATTiny85



Engineer: John Mentgen

HKN Beta Psi

Sheet: /

File: Watch_TH.kicad_sch

Title: HKN DIY Watch Project

Size: A4 Date: 2026-02-11

KiCad E.D.A. kicad 7.0.7

Rev: 1.0

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